GAN-Based Medical Image Generation for Skin Disease Classification

A Report Submitted
in Partial Fulfillment of the Requirements
for the Degree of
Mini Project (CSN17351)
in
Computer Science & Engineering

by
Mohd Shaad (20233187)
Mayukh Panja (20233534)
Sanmay Kumar Biswas (20233253)
Sahil Mukeshbhai Patel (20233556)

Section C – Team C49

to the
COMPUTER SCIENCE AND ENGINEERING DEPARTMENT
MOTILAL NEHRU NATIONAL INSTITUTE OF TECHNOLOGY
ALLAHABAD PRAYAGRAJ
October 2026

UNDERTAKING

We declare that the work presented in this report titled "GAN-Based Medical Image Generation for Skin Disease Classification", submitted to the Department of Computer Science and Engineering, Motilal Nehru National Institute of Technology Allahabad, in partial fulfillment of the requirements of the Mini Project (CSN17351), is the work carried out by our team. The report describes the experiments performed by us and acknowledges the dataset and references used. This work has not been submitted by us for the award of any other degree or qualification.

October 2026, Prayagraj

Mohd Shaad (20233187)	Mayukh Panja (20233534)
Sanmay Kumar Biswas (20233253)	Sahil Mukeshbhai Patel (20233556)

CERTIFICATE

Certified that the work contained in this report titled "GAN-Based Medical Image Generation for Skin Disease Classification", submitted by Mohd Shaad (20233187), Mayukh Panja (20233534), Sanmay Kumar Biswas (20233253), and Sahil Mukeshbhai Patel (20233556), has been carried out under the supervision of Dr. Rajitha B. for the Mini Project (CSN17351), Section C, Team C49, Department of Computer Science and Engineering, Motilal Nehru National Institute of Technology Allahabad.

October 2026, Prayagraj

(Dr. Rajitha B.)
Project Mentor
Department of Computer Science and Engineering
M.N.N.I.T. Allahabad

Preface

This report presents the work carried out for the mini project titled *GAN-Based Medical Image Generation for Skin Disease Classification* (CSN17351) by Section C, Team C49. The project investigates whether class-conditioned synthetic skin-disease images generated using a Conditional Generative Adversarial Network (CGAN) can support a downstream Convolutional Neural Network (CNN) classification task.

The report describes the dataset used, the CGAN-based image-generation workflow, the classifier experiments, the observed results, and the limitations of the study. The work is experimental and is not intended for clinical diagnosis or treatment.

Acknowledgements

We express our sincere gratitude to our project mentor, Dr. Rajitha B., for her guidance and support during this mini project. We also acknowledge the Department of Computer Science and Engineering, Motilal Nehru National Institute of Technology Allahabad, for providing the academic environment and opportunity to undertake this work.

We acknowledge the contributors of the Skin Disease Classification Image Dataset hosted on Kaggle, which was used for the experiments. All results and limitations reported in this document are based on the experiments conducted by the project team.

Contents

Preface **iv**

Acknowledgements **v**

1 Introduction **1**

1.1 Background 1

1.2 Problem Statement 1

1.3 Objectives 2

1.4 Scope 2

2 Related Work **3**

3 Proposed Work **4**

3.1 Dataset 4

3.2 Overall Workflow 5

3.3 Conditional GAN 5

3.4 CNN Classifier 5

3.5 Experimental Conditions 5

4 Experimental Setup and Results Analysis **7**

4.1 Implementation Configuration 7

4.2 Evaluation Metrics 8

4.3 Classification Results 8

4.4 Synthetic Sample Label Agreement 8

4.5 Interpretation 9

4.6 Limitations 9

5 Conclusion and Future Work **10**

A Repository and Reproducibility **11**

B Academic Disclaimer **12**

References **13**

Chapter 1

Introduction

1.1 Background

Skin conditions can exhibit substantial visual variation, while labelled medical-image datasets may be limited in size or unevenly distributed across diagnostic categories. Generative models offer a way to produce additional image samples. However, generated images are not automatically realistic, diverse, or correctly labelled. Their usefulness must therefore be assessed through sample inspection and downstream-task evaluation.

Generative Adversarial Networks (GANs) consist of a generator and a discriminator trained in an adversarial setting Goodfellow et al. [2014]. A Conditional GAN (CGAN) extends this framework by providing class or other conditioning information to the generator and discriminator Mirza and Osindero [2014]. This project applies class conditioning to skin-disease image generation.

1.2 Problem Statement

The project investigates whether synthetic skin-disease images generated by a CGAN can augment a real-image training set and improve a CNN classifier's performance across nine skin-disease classes.

1.3 Objectives

The objectives of the project are:

1. Prepare a nine-class skin-disease image dataset.
2. Train a CGAN conditioned on disease-class labels.
3. Generate 20 synthetic images per class, for 180 images in total.
4. Train CNN classifiers using real-only and real-plus-synthetic data.
5. Compare validation accuracy and macro-averaged classification metrics.
6. Document limitations and avoid unsupported clinical claims.

1.4 Scope

This work is an academic and experimental study of image generation and classification. It does not establish clinical validity and must not be used to diagnose, screen, or guide treatment for any person.

Chapter 2

Related Work

GANs were introduced as an adversarial learning framework in which a generator learns to produce samples that resemble a training distribution, while a discriminator learns to distinguish generated samples from real ones Goodfellow et al. [2014]. Conditional GANs incorporate additional information, such as class labels, into the generation process Mirza and Osindero [2014]. This provides a mechanism for requesting samples associated with a specified class.

In the present project, class conditioning is used to request synthetic images from one of nine skin-disease categories. The usefulness of the generated samples is examined through a downstream classification comparison. The project does not claim that generated images are medically validated; their visual quality and label reliability remain limitations of the experimental setup.

Chapter 3

Proposed Work

3.1 Dataset

The project used the *Skin Disease Classification Image Dataset* provided by riyaelizashaju on Kaggle riyaelizashaju [2026]. The experiments used the `Split_smol` subset. The reported working split contained 697 training images and 181 validation images. No independent test folder was available in the dataset split used.

The class-index order used in the project was:

1. Actinic keratosis
2. Atopic Dermatitis
3. Benign keratosis
4. Dermatofibroma
5. Melanocytic nevus
6. Melanoma
7. Squamous cell carcinoma
8. Tinea Ringworm Candidiasis
9. Vascular lesion

Dataset page: `https://www.kaggle.com/datasets/riyaelizashaju/skin-disease-classification-image-dataset`.

3.2 Overall Workflow

The workflow consists of dataset preparation, CGAN training, class-conditional image generation, preliminary sample/label-consistency checks, CNN classifier training under two data conditions, and validation-based comparison.

3.3 Conditional GAN

The generator receives a latent noise vector of dimension 128 together with a one-hot class vector of dimension 9 and produces a 64 by 64 RGB image. The discriminator is trained to distinguish real from generated images while using the class condition. The adversarial objective encourages generated samples to resemble the training distribution while respecting the requested class condition.

The project generated 20 samples for each of the nine classes, for 180 synthetic images in total. The generated images were observed to be blurry and to lack fine visual detail.

3.4 CNN Classifier

The downstream classifier is a compact CNN with three Conv2D blocks, global average pooling, a dense layer, dropout, and a nine-class softmax output. It contains 102,089 parameters. Both experimental conditions used the same classifier design and were trained for 10 epochs.

3.5 Experimental Conditions

Two training conditions were evaluated:

Condition	**Training data**	**Count**
Real-only	Real images	697
Real + synthetic	697 real images + 180 synthetic images	877

Both conditions were evaluated using the same validation set of 181 real images. The augmented run initially encountered an input-pipeline exhaustion warning. It was corrected using a repeated dataset and `steps_per_epoch=28`.

Chapter 4

Experimental Setup and Results Analysis

4.1 Implementation Configuration

The CGAN notebook used the following configuration:

Parameter	**Value**
Image dimensions	64×64 pixels
Image channels	3 (RGB)
Batch size	64
Latent dimension	128
Number of classes	9
Training epochs	200
Learning rate	0.0002
Random seed	42
Sample generation interval	Every 5 epochs

The CNN classifier architecture consists of three convolutional blocks, global average pooling, a dense layer, dropout, and a nine-class softmax output. The classifier contains 102,089 parameters and was trained for 10 epochs in each condition.

4.2 Evaluation Metrics

Final validation accuracy is the fraction of validation examples classified correctly at the end of a run. Macro F1-score is the unweighted mean of class-wise F1-scores. Macro precision and recall similarly average class-level precision and recall. No confidence intervals or repeated-run estimates are available.

4.3 Classification Results

The final validation results are summarized in the comparison table below.

Metric	**Real-only**	**Real + synthetic**
Training images	697	877
Final validation accuracy	29.28%	28.73%
Macro F1-score	0.2223	0.2315
Best validation accuracy	32.60% (epoch 6)	31.49% (epoch 7)
Final validation loss	1.6967	1.7739

The real-only classifier achieved 29.28% final validation accuracy and a macro F1-score of 0.2223. The real-plus-synthetic classifier achieved 28.73% final validation accuracy and a macro F1-score of 0.2315.

4.4 Synthetic Sample Label Agreement

A preliminary evaluation compared the requested labels of generated images with predictions from the baseline classifier. The classifier agreed with the intended labels for 38 of 180 generated images (21.11%). This is only a weak proxy, not an independent ground-truth assessment. Because the baseline classifier performed poorly, disagreement is not definitive proof that a generated image has the wrong class.

4.5 Interpretation

Final validation accuracy decreased by 0.55 percentage points after adding synthetic images, while macro F1 increased by 0.0092. The results do not support a claim that synthetic augmentation improved overall validation accuracy. Generated samples were visually blurry and lacked fine visual detail.

The saved classifier files represent final-epoch weights, not best-validation-epoch checkpoints. Only one run per condition was evaluated, so no statistical significance analysis or repeated-run estimate is available.

4.6 Limitations

The main limitations of the study are:

- The dataset was small, with 697 training and 181 validation images.
- No independent test set was available.
- Generated 64 by 64 images were visibly blurry.
- Only one run per condition was evaluated; no statistical significance analysis was performed.
- Label agreement relied on a weak baseline classifier, not expert annotation.
- Saved classifier weights are final-epoch rather than best-validation checkpoints.
- No external clinical validation or dermatologist review was conducted.

Chapter 5

Conclusion and Future Work

This project implemented a class-conditioned GAN pipeline and evaluated adding synthetic images to a CNN classifier's training set. The real-plus-synthetic condition did not improve final validation accuracy (28.73% versus 29.28% for real-only), although macro F1 increased slightly (0.2315 versus 0.2223). Given the dataset size, blurry samples, weak label-agreement proxy, and lack of an independent test set, the evidence is insufficient to conclude that synthetic images provide a dependable benefit.

Future work should improve image resolution and generator stability, obtain expert review of generated samples, use a larger and more diverse dataset, preserve an independent test set, save best-validation checkpoints, and repeat experiments across multiple random seeds.

Appendix A

Repository and Reproducibility

The project repository contains notebooks, trained generator and classifier models, and experiment result files. The intended workflow is to obtain the dataset from Kaggle, prepare the expected class-folder structure, run the CGAN notebook, generate synthetic samples, and run the evaluation pipeline. Exact package versions, runtime settings, and paths should be verified against the notebooks before claiming full environment reproducibility.

The project GitHub repository is: `https://github.com/SanmayKB/C-49-GAN-based-model-for-generation-of-images-for-disease-classification`.

Appendix B

Academic Disclaimer

This work is an academic mini-project. The models and outputs have not been clinically validated and are not intended to diagnose, screen, or guide treatment for any person. Performance values are limited to the stated experimental split and setup.

References

I. Goodfellow et al., "Generative Adversarial Nets," in *Advances in Neural Information Processing Systems*, 2014.

M. Mirza and S. Osindero, "Conditional Generative Adversarial Nets," arXiv:1411.1784, 2014.

riyaelizashaju, "Skin Disease Classification Image Dataset," Kaggle. Available: `https://www.kaggle.com/datasets/riyaelizashaju/skin-disease-classification-image-dataset`.